

Ravi Kumar

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PROFILE

Senior **Data & AI Engineer** with **7+ years of experience** designing scalable **AI, GenAI, and cloud-native data platforms** across healthcare, gaming, media, and enterprise domains. Experienced in building production-grade **LLM applications, Agentic AI workflows, RAG systems, AI observability platforms, and ML-powered analytics solutions** using **LangChain, LangGraph, AWS, Databricks, Python, and modern AI infrastructure**. Strong expertise in **LLMOps/MLOps, scalable data engineering, AI deployment, cloud architecture, and enterprise AI platform development** with hands-on experience in deploying intelligent systems on AWS and Databricks ecosystems.

TECHNICAL SKILLS

Programming & Frameworks: Python, SQL, PySpark, FastAPI

AI/ML & GenAI: LLMs, RAG, Agentic AI, LangChain, LangGraph, NLP, Scikit-learn, XGBoost, MLflow, OpenCV, Hugging Face, Prompt Engineering, MCP (Model Context Protocol)

Cloud & AI Infrastructure: AWS (Bedrock, SageMaker, EKS, ECR, EC2, Lambda, S3, Glue, Athena, CloudWatch, Kinesis, MSK), Databricks, Snowflake

MLOps & LLMOps: Docker, Kubernetes, GitHub Actions, Terraform, CI/CD, Model Deployment, Observability, Prompt Versioning

Data Engineering & Streaming: Spark, Delta Lake, Airflow, Kafka, Kinesis, Medallion Architecture, Data Warehousing, ETL Pipelines

Vector & Search Technologies: FAISS, Pinecone, Hybrid Search, Semantic Search, Embedding Models, Reranking

EXPERIENCE

ConcertAI

APR 2023 – Present

Data Engineer

Bangalore, India

- Led the migration of legacy **AWS Glue** and **Snowflake** pipelines to the **Databricks Lakehouse** platform, implementing **Medallion architecture** on **Delta Lake** with **Unity Catalog** for centralized governance, scalable processing, and enterprise data management.
- Designed and deployed enterprise-scale **Agentic RAG** knowledge assistants integrating **Confluence, GitHub**, and internal documentation using **LangChain, LangGraph, FAISS/Pinecone vector databases, hybrid retrieval, conversational memory**, and **tool-calling agents** to enable context-aware reasoning, intelligent knowledge retrieval, and autonomous workflow assistance, reducing onboarding time by **50%**.
- Designed and developed **AI-powered observability** and **self-healing platforms** for **AWS Glue** and **Databricks** pipelines using **Amazon Bedrock, AWS Lambda, CloudWatch, Spark metadata analysis**, and **RAG-based incident intelligence** to automate anomaly detection, root cause analysis, and remediation recommendations for enterprise-scale data pipelines.
- Built **Agentic AI-driven incident management workflows** using **LangChain, LangGraph, LLMs, Jira**, and **Slack integrations** to automate incident classification, severity prioritization, intelligent team routing, RCA generation, and operational notifications for enterprise data platforms.
- Built **MCP (Model Context Protocol)-based integrations** and **tool orchestration frameworks** for AI agents to securely interact with enterprise systems, APIs, databases, and business applications, enabling context-aware reasoning, dynamic tool invocation, and autonomous task execution across **Agentic AI workflows**.
- Architected and implemented production-grade **LLMOps** and **CI/CD workflows** for **GenAI** and **Agentic AI** applications using **Docker, FastAPI, GitHub Actions**, and **AWS services (EKS, ECR, EC2, Lambda, S3)** to support scalable model deployment, inference serving, observability, and enterprise-scale **LLM lifecycle management**.

- Leveraged **Claude** and **Generative AI**-assisted development workflows to accelerate **test data generation**, **code migration**, **requirement analysis**, and **document generation** tasks, reducing manual effort by up to **60%** and significantly improving engineering productivity.

Agilisium Consulting

Dec 2021 – Mar 2023

Data Engineer

Bangalore, India

- Designed distributed transformation pipelines on **Databricks** using **PySpark** to process and normalize large-scale semi-structured datasets from **S3**, loading curated datasets into **Snowflake** for downstream analytics and dashboarding, while orchestrating ingestion-to-warehouse workflows using **Airflow DAGs**.
- Developed **machine learning solutions** including **churn prediction**, **player segmentation**, and **retention analytics** using **Python**, **SQL**, **Scikit-learn**, and **XGBoost**, leveraging behavioral and transactional data to drive personalized engagement, user targeting, and predictive business insights.
- Designed and implemented scalable **ML architecture** and **MLOps pipelines** on **AWS** and **Databricks** using **S3**, **Lambda**, **Glue**, **Athena**, **Docker**, **MLflow**, and **Amazon SageMaker** for feature engineering, experiment tracking, model training, deployment, monitoring, and real-time inference workflows across AI applications.
- Automated provisioning and deployment of **AI/ML infrastructure** and **Databricks resources** using **Terraform**, **AWS CodeCommit**, **CodeBuild**, **CodePipeline**, and **Git integration**, enabling **Infrastructure as Code (IaC)**, **CI/CD automation**, version-controlled releases, and standardized cloud-based ML platform management.

LinkCXO Global Private Ltd

Apr 2019 – June 2021

Product Engineer

Bangalore, India

- Developed an **AI-powered conversational chatbot** using **Rasa**, **Python**, and **NLP-based intent recognition** and **dialogue management** to provide users with real-time access to industry reports, application guidance, and platform support, improving user engagement and self-service capabilities.
- Built an **aspect-based sentiment analysis pipeline** using **Python**, **Pandas**, **NLTK**, **spaCy**, **Scikit-learn**, and machine learning techniques to extract sentiment for specific business attributes from customer reviews and feedback data, enabling actionable business insights and analytics.
- Developed a **recommendation system** using **Python**, **Scikit-learn**, **collaborative filtering**, and behavioral analytics to deliver personalized content recommendations and suggest similar users based on likes, comments, views, and user interaction patterns, improving user engagement and platform connectivity.
- Developed an **image similarity** and **facial matching system** using **OpenCV**, **computer vision**, **deep learning**, and machine learning techniques to identify visually similar faces and generate compatibility scores for user recommendations within a dating application.
- Engineered scalable cloud deployment workflows for **NLP**, **recommendation**, and **computer vision models** using **Docker**, **FastAPI**, **AWS EC2**, **Lambda**, **S3**, and **GitHub Actions** for production-grade AI applications.

EDUCATION

Visvesvaraya Technological University, Belgaum

Bachelor of Engineering in Computer Science

Dharwad, Karnataka

Aug 2014 – May 2018